

Shell Scripting

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Lab-2

1. Create a script that asks for user name then send a greeting to him.

```
q.sh
1  #!/usr/bin/bash
2
3  echo What's your name
4
5  read username
6
7  echo "Happy to meet you $username"
8
```

2. Create a script called s1 that calls another script s2 where:

- In s1 there is a variable called x, it's value 5
- Try to print the value of x in s2 by two different ways.

```
s1.sh
1  #!/usr/bin/bash
2
3  export x=5
4
5  ./s2.sh $x
6
```

```
s2.sh
1  #!/usr/bin/bash
2
3  echo "Value via export: $x"
4
5  echo "Value via argument: $1"
6
```

3. Create a script called mycp where:

- It copies a file to another
- It copies multiple files to a directory.

```
mycp.sh
1  #!/usr/bin/bash
2
3  cp -r "$@"
4
```

4. Create a script called mycd where:

- It changed directory to the user home directory, if it is called without arguments.
- Otherwise, it change directory to the given directory.

```
mycd.sh
1  #!/usr/bin/bash
2
3  if [ -z $1 ]; then
4  | cd ~
5  else
6  | cd $1
7  fi
8
```

5. Create a script called myls where:

a. It lists the current directory, if it is called without arguments.

b. Otherwise, it lists the given directory.

a. -l: list in long format

b. -a: list all entries including the hiding files.

c. -d: if an argument is a directory, list only its name

d. -i: print inode number

e. -R: recursively list subdirectories

mysls -option1 -option2

mysls -option2 -option1

mysls -option1option2

mysls -option2option1

```
mysls.sh
1  #!/usr/bin/bash
2
3  options=""
4  while getopts "laidR" opt; do
5      case $opt in
6          l) options="$options -l" ;;
7          a) options="$options -a" ;;
8          i) options="$options -i" ;;
9          d) options="$options -d" ;;
10         R) options="$options -R" ;;
11         *) echo "Invalid option"; exit 1 ;;
12     esac
13 done
14
15 shift $((OPTIND-1))
16
17 dir=${1:-.}
18
19 ls $options "$dir"
```

7. Create a script called mytest where:

a. It check the type of the given argument (file/directory)

b. It check the permissions of the given argument (read/write/execute)

```
mytest.sh
1  #!/usr/bin/bash
2
3  target=$1
4
5  if [ ! -e "$target" ]; then
6      echo "Path does not exist."
7      exit 1
8  fi
9
10 [ -f "$target" ] && echo "$target is a File"
11 [ -d "$target" ] && echo "$target is a Directory"
12
13 [ -r "$target" ] && echo "Read: Yes" || echo "Read: No"
14 [ -w "$target" ] && echo "Write: Yes" || echo "Write: No"
15 [ -x "$target" ] && echo "Execute: Yes" || echo "Execute: No"
16
```